Taller 03

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Métodos Numéricos

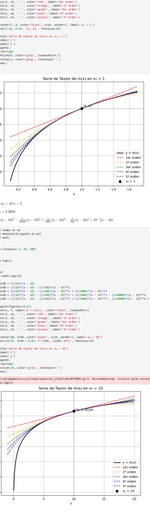
Ejercicio 1

- Grafique los conos de las series de Taylor de series Ordenes para los siguientes casos:

1. $\ln(x)$ $x_0 = 1$ $n = 5$

$$\left\{(x-1)^1, \frac{1}{2}(x-1)^2, \frac{1}{6}(x-1)^3, \frac{1}{12}(x-1)^4, \frac{1}{20}(x-1)^5\right\}$$

```
import numpy as np
import matplotlib.pyplot as plt
import math

x = np.linspace(0.5, 3.5, 400)

f = np.ln(x)

y0 = 0
y1 = 0 + (x - 1)*f(1)
y2 = 0 + (x - 1)*f(1) + 1/2 * (x - 1)**2 * f(2)
y3 = 0 + (x - 1)*f(1) + 1/2 * (x - 1)**2 * f(2) + 1/6 * (x - 1)**3 * f(3)
y4 = 0 + (x - 1)*f(1) + 1/2 * (x - 1)**2 * f(2) + 1/6 * (x - 1)**3 * f(3) + 1/24 * (x - 1)**4 * f(4)
y5 = 0 + (x - 1)*f(1) + 1/2 * (x - 1)**2 * f(2) + 1/6 * (x - 1)**3 * f(3) + 1/24 * (x - 1)**4 * f(4) + 1/120 * (x - 1)**5 * f(5)

plt.figure(figsize=(8,6))
plt.plot(x, f, label='ln(x)', color='black', linewidth=3)
plt.plot(x, y0, '--', color='red', label='Cero orden')
plt.plot(x, y1, '--', color='orange', label='1er orden')
plt.plot(x, y2, '--', color='green', label='2do orden')
plt.plot(x, y3, '--', color='blue', label='3er orden')
plt.plot(x, y4, '--', color='purple', label='4to orden')
plt.plot(x, y5, '--', color='purple', label='5to orden')

plt.scatter(1, 0, color='black', s=60, zorder=5, label='$x_0 = 1$')
plt.text(1.0, 0.2, r'$(1, 0)$', fontsize=10)

plt.title('Serie de Taylor de ln(x) en $x_0 = 1$')
plt.xlabel('$x$')
plt.ylabel('$y$')
plt.legend()
plt.grid(True)
plt.axhline(0, color='gray', linewidth=0.5)
plt.axvline(1, color='gray', linewidth=0.5)
plt.show()
```



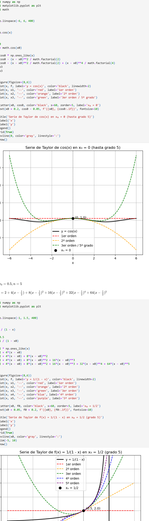
$\ln(x)$ $x_0 = 10$ $n = 5$

$\ln(10) = 2.3025$

$\frac{1}{10}(x-10)^1 - \frac{1}{200}(x-10)^2 + \frac{1}{6000}(x-10)^3 - \frac{1}{240000}(x-10)^4 + \frac{1}{7200000}(x-10)^5$

```
import numpy as np
import matplotlib.pyplot as plt
import math

x = np.linspace(1, 20, 200)

f = np.ln(x)

y0 = 0
y1 = 0 + (x - 10)*f(10) + (x - 10)*(1/10)
y2 = 0 + (x - 10)*f(10) + (x - 10)*(1/10) + (x - 10)**2 * (-1/200)
y3 = 0 + (x - 10)*f(10) + (x - 10)*(1/10) + (x - 10)**2 * (-1/200) + (x - 10)**3 * (1/6000)
y4 = 0 + (x - 10)*f(10) + (x - 10)*(1/10) + (x - 10)**2 * (-1/200) + (x - 10)**3 * (1/6000) + (x - 10)**4 * (-1/240000)
y5 = 0 + (x - 10)*f(10) + (x - 10)*(1/10) + (x - 10)**2 * (-1/200) + (x - 10)**3 * (1/6000) + (x - 10)**4 * (-1/240000) + (x - 10)**5 * (1/7200000)

plt.figure(figsize=(8,6))
plt.plot(x, f, label='ln(x)', color='black', linewidth=3)
plt.plot(x, y0, '--', color='red', label='Cero orden')
plt.plot(x, y1, '--', color='orange', label='1er orden')
plt.plot(x, y2, '--', color='green', label='2do orden')
plt.plot(x, y3, '--', color='blue', label='3er orden')
plt.plot(x, y4, '--', color='purple', label='4to orden')
plt.plot(x, y5, '--', color='purple', label='5to orden')

plt.scatter(10, 0.0, color='black', s=60, zorder=5, label='$x_0 = 10$')
plt.text(10.0, 0.2, r'$(10, 0)$', fontsize=10)

plt.title('Serie de Taylor de ln(x) en $x_0 = 10$')
plt.xlabel('$x$')
plt.ylabel('$y$')
plt.legend()
plt.grid(True)
plt.axhline(0, color='gray', linewidth=0.5)
plt.axvline(10, color='gray', linewidth=0.5)
plt.show()
```

```
C:\Users\Ing\Desktop\Luis\Tmp\Squarespace_Client\ANIMATIONS.py:5: PyrtleShelling: Invalid value encountered in log
  f = np.ln(x)
```



$\cos(x)$ $x_0 = 0$ $n = 5$

$\frac{x^2}{2} - \frac{x^4}{24}$

```
import numpy as np
import matplotlib.pyplot as plt
import math

x = np.linspace(-3, 3, 400)

f = np.cos(x)

y0 = 1
y1 = 1 - x**2/2
y2 = 1 - x**2/2 + x**4/24
y3 = 1 - x**2/2 + x**4/24 - x**6/720
y4 = 1 - x**2/2 + x**4/24 - x**6/720 + x**8/40320
y5 = 1 - x**2/2 + x**4/24 - x**6/720 + x**8/40320 - x**10/3628800

plt.figure(figsize=(8,6))
plt.plot(x, f, label='$\cos(x)$', color='black', linewidth=3)
plt.plot(x, y0, '--', color='red', label='Cero orden')
plt.plot(x, y1, '--', color='orange', label='1er orden')
plt.plot(x, y2, '--', color='green', label='2do orden')
plt.plot(x, y3, '--', color='blue', label='3er orden')
plt.plot(x, y4, '--', color='purple', label='4to orden')

plt.scatter(0, 1, color='black', s=60, zorder=5, label='$x_0 = 0$')
plt.text(0.0, 1.2, r'$(0, 1)$', fontsize=10)

plt.title('Serie de Taylor de $\cos(x)$ en $x_0 = 0$ (hasta grado 5)')
plt.xlabel('$x$')
plt.ylabel('$y$')
plt.legend()
plt.grid(True)
plt.axhline(0, color='gray', linewidth=0.5)
plt.show()
```

$f(x, y) = 0.5 \cdot x + 1$

$P_1(x) = 1 + 0(x - \frac{1}{2}) + 0(x - \frac{1}{2})^2 + 0(x - \frac{1}{2})^3 + 0(x - \frac{1}{2})^4$

```
import numpy as np
import matplotlib.pyplot as plt

x = np.linspace(-3, 3.5, 400)

f = 1 / (1 - x)

y0 = 1
y1 = 1 + x
y2 = 1 + x + x**2
y3 = 1 + x + x**2 + x**3
y4 = 1 + x + x**2 + x**3 + x**4
y5 = 1 + x + x**2 + x**3 + x**4 + x**5

plt.figure(figsize=(8,6))
plt.plot(x, f, label='$\frac{1}{1-x}$', color='black', linewidth=3)
plt.plot(x, y0, '--', color='red', label='Cero orden')
plt.plot(x, y1, '--', color='orange', label='1er orden')
plt.plot(x, y2, '--', color='green', label='2do orden')
plt.plot(x, y3, '--', color='blue', label='3er orden')
plt.plot(x, y4, '--', color='purple', label='4to orden')

plt.scatter(0, 1, color='black', s=60, zorder=5, label='$x_0 = 0$')
plt.text(0.0, 1.2, r'$(0, 1)$', fontsize=10)

plt.title('Serie de Taylor de $f(x) = 1/(1 - x)$ en $x_0 = 1/2$ (grado 5)')
plt.xlabel('$x$')
plt.ylabel('$y$')
plt.legend()
plt.grid(True)
plt.axhline(0, color='gray', linewidth=0.5)
plt.ylim(-1, 10)
plt.show()
```

Ejercicio 2

Encuentre el polinomio de Lagrange para los siguientes datos y grafique

1. $(0, 0), (8), \left(\frac{1}{2}, 1\right), (9, 1)$

$x_0 = 0, \quad x_1 = 36, \quad x_2 = 49, \quad x_3 = 50$

$y_0 = 0, \quad y_1 = 0.5, \quad y_2 = \frac{1}{2}, \quad y_3 = 1$

Bases de Lagrange

$L_0(x)$

$$L_0(x) = \frac{(x-36)(x-49)(x-50)}{(0-36)(0-49)(0-50)} = \frac{(x-36)(x-49)(x-50)}{-245000}$$

$L_1(x)$

$$L_1(x) = \frac{x(x-49)(x-50)}{(36-0)(36-49)(36-50)} = \frac{x(x-49)(x-50)}{14000}$$

$L_2(x)$

$$L_2(x) = \frac{x(x-36)(x-50)}{(49-0)(49-36)(49-50)} = \frac{x(x-36)(x-50)}{-26000}$$

$L_3(x)$

$$L_3(x) = \frac{x(x-36)(x-49)}{(50-0)(50-36)(50-49)} = \frac{x(x-36)(x-49)}{35000}$$

Polinomio

$$P(x) = 0L_0(x) + \frac{1}{2}L_1(x) + \frac{1}{2}L_2(x) + 1 \cdot L_3(x)$$

$$P(x) = \left(\frac{x(x-36)(x-49)}{28000}\right) - \left(\frac{x(x-36)(x-49)}{14000}\right) + \left(\frac{x(x-36)(x-50)}{26000}\right) + \left(\frac{x(x-36)(x-49)}{35000}\right)$$

$$P(x) = -0.00x^3 + 10^{-2}x^2 - 1.86x + 10^{-2}x^3 + 0.0076x$$

```
import numpy as np
import matplotlib.pyplot as plt
import math

np = np.array([0, 36, 49, 50])
fp = np.array([0, 0.5, 0.5, 1])

x = np.linspace(0, 50, 400)

L0 = (x-36)*(x-49)*(x-50) / -245000
L1 = x*(x-49)*(x-50) / 14000
L2 = x*(x-36)*(x-50) / -26000
L3 = x*(x-36)*(x-49) / 35000

P = fp[0]*L0 + fp[1]*L1 + fp[2]*L2 + fp[3]*L3

plt.figure(figsize=(8,6))
plt.plot(x, P, label='Polinomio de Lagrange', color='black', linewidth=3)
plt.scatter(np.nan, np.nan)  # para que aparezca en la leyenda
for i, (xi, yi) in enumerate(zrip(np, fp), 1):
    plt.text(xi, yi+0.05, f'P({xi:.0f})')
plt.xlabel('$x$')
plt.ylabel('$P(x)$')
plt.title('Interpolación de Lagrange')
plt.legend()
plt.grid(True)
plt.show()
```

$(1, 5), (3, 7), (5, 11), (7, 16)$

$x_0 = 1, \quad x_1 = 3, \quad x_2 = 5, \quad x_3 = 7$

$y_0 = 5, \quad y_1 = 7, \quad y_2 = 11, \quad y_3 = 16$

Bases de Lagrange

$L_0(x)$

$$L_0(x) = \frac{(x-3)(x-5)(x-7)}{(1-3)(1-5)(1-7)} = \frac{(x-3)(x-5)(x-7)}{-48}$$

$L_1(x)$

$$L_1(x) = \frac{(x-1)(x-5)(x-7)}{(3-1)(3-5)(3-7)} = \frac{(x-1)(x-5)(x-7)}{8}$$

$L_2(x)$

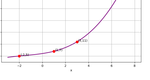
$$L_2(x) = \frac{(x-1)(x-3)(x-7)}{(5-1)(5-3)(5-7)} = \frac{(x-1)(x-3)(x-7)}{-16}$$

Polinomio

$$P(x) = 5 \cdot L_0(x) + 7 \cdot L_1(x) + 11 \cdot L_2(x)$$

$$P(x) = \frac{5}{-48}x^3 + \frac{7}{8}x^3 + 5$$

```
import numpy as np
import matplotlib.pyplot as plt
import math

x = np.linspace(0, 10, 200)

f = 0.5*x**3 + 2.5*x - 1

np = np.array([1, 3, 5])
fp = np.array([5, 7, 11])

plt.figure(figsize=(8,6))
plt.plot(x, f, label='Polinomio de Lagrange', color='BTBB00', linewidth=2)
plt.scatter(np, fp, color='red', label='Puntos dados', zorder=5)
plt.xlabel('$x$')
plt.ylabel('$P(x)$')
plt.title('Interpolación de Lagrange')
plt.legend()
plt.grid(True)
plt.show()
```



$(-3, 5), (1, 7), (3, 11), (7, 34)$

$x_0 = -3, \quad x_1 = 1, \quad x_2 = 3, \quad x_3 = 7$

$y_0 = 5, \quad y_1 = 7, \quad y_2 = 11, \quad y_3 = 34$

Bases de Lagrange

$L_0(x)$

$$L_0(x) = \frac{(x-1)(x-3)(x-7)}{(-3-1)(-3-3)(-3-7)} = \frac{(x-1)(x-3)(x-7)}{-240}$$

$L_1(x)$

$$L_1(x) = \frac{(x+3)(x-3)(x-7)}{(1+3)(1-3)(1-7)} = \frac{(x+3)(x-3)(x-7)}{48}$$

$L_2(x)$

$$L_2(x) = \frac{(x+3)(x-1)(x-7)}{(3+3)(3-1)(3-7)} = \frac{(x+3)(x-1)(x-7)}{-48}$$

$L_3(x)$

$$L_3(x) = \frac{(x+3)(x-1)(x-3)}{(7+3)(7-1)(7-3)} = \frac{(x+3)(x-1)(x-3)}{240}$$

Polinomio

$$P(x) = \frac{-10}{-240} + \frac{35}{48} + \frac{55}{-48} + \frac{187}{240}$$

$$P(x) = 0.0004x^3 + 0.0727x^2 + 0.7292x + 0.0000$$

```
import numpy as np
import matplotlib.pyplot as plt
import math

np = np.array([-3, 1, 3, 7])
fp = np.array([5, 7, 11, 34])

x = np.linspace(-3, 7, 400)

P = 10/240*x**3 + 35/48*x**2 + 55/48*x + 187/240

plt.figure(figsize=(8,6))
plt.plot(x, P, label='Polinomio de Lagrange', color='purple', linewidth=2)
plt.scatter(np, fp, color='red', zorder=5, label='Puntos dados')
for xi, yi in zrip(np, fp):
    plt.text(xi, yi+0.5, f'P({xi:.0f})')
plt.xlabel('$x$')
plt.ylabel('$P(x)$')
plt.title('Interpolación de Lagrange')
plt.legend()
plt.grid(True)
plt.show()
```